Machine Learning
Final Project: Predicting Credit Card Default Risk
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1. Problem Description and Business Relevance

For banks and fintech lenders, credit card default poses a significant risk since late payments result in monetary losses and higher operating expenses. A trustworthy default prediction model can help institutions prioritize early interventions for high-risk clients, reduce overall credit losses, and make more informed lending decisions (e.g., limit increases, credit approvals). Using demographic data, credit limit, repayment status history, bill statements, and past payments, this project frames default prediction as a binary classification task: forecasting whether a customer will default next month.

Machine learning is suitable because default risk is dependent on several interrelated factors (e.g., late payments combined with rising bills or low repayments). It can learn nonlinear patterns and interactions beyond simple rules while still producing risk scores based on probability.

2. Description of the Dataset

The Default of Credit Card Clients dataset (UCI) is used in this project. The target variable is the next month's default payment (1 = default, 0 = no default), and each row represents a single credit card customer.

- The primary feature groups of the 30,000 records in the dataset are:
  - Demographics: age, marital status, education, and sex
  - LIMIT_BAL is the credit limit.
  - History of repayment status: PAY_0–PAY_6
  - Bill statements: BILL_AMT1–BILL_AMT6;
  - Previous payments: PAY_AMT1–PAY_AMT6
- Distribution in class :
  - Non-default (0): 23,364
  - 6,636 is the default (1).
  - 0.2212 (≈ 22.12%) is the default rate.

This suggests that the dataset is somewhat unbalanced, so evaluation should concentrate on metrics other than accuracy (particularly for the default class).

3. Features engineering choices

Cleaning

- Missing value check: No missing values were detected.
- Removed ID: the ID column only serves as an identifier and has no predictive value, it was eliminated. Maintaining it could lead to erroneous patterns or mislead the model.

The dataset contains both numeric and categorical variables. Even though SEX, EDUCATION, and MARRIAGE are stored as integers, they are categorical labels, not meaningful numeric quantities. Therefore:

- Categorical variables were one-hot encoded.
- Numeric variables were handled differently depending on the model:
  - Logistic Regression: numeric features were standardised (scaling helps linear models converge and prevents large-scale features dominating).
  - Gradient Boosting (tree model): numeric features were not scaled (tree models are scale-invariant).

Data leakage is prevented and preprocessing is applied uniformly to training and test data by using Pipeline and ColumnTransformer.
The dataset's current features—repayment delays, bills, and payment amounts—are the main tools used in this project because they already contain strong predictive signals.
Without imposing false numerical values, one-hot encoding converts categorical variables into a format appropriate for modelling.

4. Model Development (Primary + Comparison)

4.1 Comparison Model: Logistic Regression (Baseline)

Logistic regression was selected as the baseline, because of its ease of use, interpretability, and popularity in risk scoring. Class_weight ("balanced") was used to lessen bias toward the majority class because the dataset is unbalanced.

4.2 Primary Model: Gradient Boosting

Gradient Boosting model was selected as the main model, because it captures the nonlinear relationships and feature interactions typical of financial risk data. To address class imbalance, sample weights were calculated and passed during training because this model does not directly accept class_weight.

5. Results and Evaluation

Because this is an imbalanced classification problem, performance was evaluated using:

- **ROC-AUC** (ranking quality across thresholds)
- **Precision, Recall, and F1-score** (focused on default class = 1)
- **Confusion matrix**
- **ROC curve**

5.1 Baseline: Logistic Regression — Results

- **ROC-AUC: 0.71**
- **Default class (1):**
  - Precision: **0.3677**
  - Recall: **0.6323**
  - F1-score: **0.4649**
- **Accuracy: 0.6782**

Interpretation:
The Logistic Regression model helps reduce missed high-risk customers by identifying a reasonable proportion of defaulters (recall ≈ 0.63). Nevertheless, default precision is low (≈ 0.37), indicating that a large number of customers who were identified as high-risk are actually non-defaults. This could result in an excessive number of false alarms in a real lending setting, as well as possibly limiting or rejecting clients who would have made repayments.

Confusion matrix (derived from test results, 6,000 cases):

- TN = **3230**
- FP = **1443**
- FN = **488**
- TP = **839**

So the baseline catches many defaulters (high TP), but also produces many false positives (high FP), which reduces precision.

5.2 Primary: Gradient Boosting — Results

- **ROC-AUC: 0.78**
- **Default class (1):**

- Precision: **0.4794**
- Recall: **0.6134**
- F1-score: **0.5382**

- **Accuracy: 0.7672**

Interpretation:

The Gradient Boosting model ranks risky customers more successfully across thresholds, significantly improving overall discrimination (ROC-AUC rises from 0.71 to 0.78). Additionally, it raises default precision from 0.3677 to 0.4794, resulting in fewer non-defaulters being mistakenly reported as defaulters. Although recall slightly drops (0.6323 → 0.6134), the increased precision results in a stronger overall balance (higher F1-score).

Confusion matrix (derived from test results, 6,000 cases):

- TN = **3789**
- FP = **884**
- FN = **513**
- TP = **814**

Compared with Logistic Regression, the tree model reduces false positives substantially (1443 → 884), which is a major practical improvement because it reduces unnecessary interventions for customers who would have repaid.

5.3 Model Comparison Summary

Model	ROC-AUC	Precision (Default=1)	Recall (Default=1)	F1 (Default=1)
Logistic Regression	**0.71**	0.3677	**0.6323**	0.4649
Gradient Boosting	**0.78**	**0.4794**	0.6134	**0.5382**

Conclusion from results:

Because it raises ROC-AUC and enhances precision and F1-score for the default class, gradient boosting is the superior primary model. This is beneficial for the financial industry since it increases the model's accuracy in identifying customers who pose a real risk while lowering false alarms.

6. SHAP Explainability (Global + Local)

To support transparency, SHAP was applied to the primary Gradient Boosting model. The three SHAP plots provide complementary explanations:

Global explanation (SHAP summary plot)

Features are ranked by importance for every customer in the SHAP summary plot. Repayment status variables, particularly PAY_0, hold the top spots in this model. Features like LIMIT_BAL, BILL_AMT1, and recent payment amounts (PAY_AMT1–PAY_AMT3) come next. This is in line with financial intuition: repayment patterns and recent delinquency are powerful indicators of default risk for the following month. The plot shows that while higher credit limits frequently push predictions toward non-default (negative SHAP values), indicating lower risk capacity, higher PAY_0 values (worse repayment status/delays) push predictions toward default (positive SHAP values).

Local explanation (SHAP waterfall plot)

One customer prediction is explained by the SHAP waterfall plot. The model output in the example is roughly 0.634, which is a comparatively high predicted default probability. The plot displays the variables that increased or decreased risk. For instance, certain characteristics (like favorable repayment status in a recent month) lower risk, while the customer's LIMIT_BAL and repayment-related variables increase risk. This offers a decision-level justification for the model's higher risk classification of this person.

Feature behaviour (SHAP dependence plot)

The SHAP dependence plot for PAY_0 shows how changes in repayment status affect predicted risk across many customers. As PAY_0 increases, SHAP values increase, meaning default risk rises. The colour gradient (e.g., BILL_AMT1) indicates interaction effects: the impact of late repayment can be stronger when bill amounts are higher, showing the model is learning combined financial stress patterns rather than relying on one variable alone.

Summary

Together, the results show that:

- The model’s strongest drivers are financially sensible (delinquency + repayment behaviour).
- Predictions can be explained both globally and per-customer.
- The model captures meaningful interactions (e.g., repayment delays + high bills), supporting trust and accountability.

7. Short Reflection and Limitations

- **Generalisation:** the dataset is from Taiwan (2005), so patterns may differ in other regions or modern credit systems.

- **Feature scope:** important factors like income, employment status, and macroeconomic conditions are not included.
- **Threshold choice:** the operational definition of “high risk” depends on the selected threshold; different business goals require different thresholds.

Probability calibration, cost-based threshold optimisation, and cross-group fairness testing are possible future enhancements. In conclusion, this project effectively used UCI data to create a machine learning solution for credit card default prediction. While Logistic Regression offered a clear baseline, the Gradient Boosting model performed better, maintaining strong recall while increasing ROC-AUC (0.7791) and default class precision (0.4794). SHAP explainability verified that repayment status and recent payment behaviour are the main factors influencing predictions, promoting openness and confidence.

All things considered, Gradient Boosting is the recommended model for implementation, as it strikes a better balance between identifying risky clients and minimizing false alerts, which is consistent with responsible and effective credit risk management.